

Robocar With Wireless Steering



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We have seen many projects where a robot is controlled by a remote controller. Here we describe a robot that can be steered in any direction just like you drive your car. Wireless steering senses the motion and transmits corresponding instruction to the robot through RF communication. The robot also has an obstacle detection and avoidance system implemented. Fig. 1 shows prototypes of the wireless steering and the controlled robot.

Circuit and working

Fig. 2 shows the block diagram of the complete project. The movement of the steering is sensed by the accelerometer. The data from the accelerometer is processed by Arduino Uno board and



Fig. 1: Prototypes of the wireless steering and the robot

PARTS LIST

Semiconductors:

IC1	- HT12E encoder
IC2, IC7	- 7805 5V regulator
IC3	- HT12D decoder
IC4	- PIC16F877A microcontroller
IC5	- LM324 operational amplifier
IC6	- L293D motor driver
IRTX1, IRTX2	- IR transmitter
IRRX1, IRRX2	- IR receiver
LED1-LED5	- 5mm LED
D1, D2	- 1N4007 rectifier diode
RF TX	- ASK transmitter module
RF RX	- ASK receiver module
BOARD1	- Arduino UNO board
	- Accelerometer module

Resistors (all 1/4-watt, ± 5 per cent carbon):

R1	- 750-kilo-ohm
R2, R5, R6,	
R11, R12	- 330-ohm
R3	- 33-kilo-ohm
R4, R9, R10	- 10-kilo-ohm
R7, R8	- 100-ohm
VR1, VR2	- 10-kilo-ohm preset

Capacitors:

C1-C3	- 0.1 μ F ceramic disk
C4, C5	- 22pF ceramic disk

Miscellaneous:

DIP1, DIP2	- DIP switch
S1, S2	- Push-to-on switch
M1, M2	- 12V, 100-rpm geared motor
BATT.1	- 9V battery (PP3)
BATT.2	- 12V, 4.5Ah battery

TABLE I Technical Specifications of Arduino Uno Board

Features	Specifications
Microcontroller	ATmega328
Operating voltage	5V
Input voltage (recommended)	7-12V
Input voltage (limits)	6-20V
Digital I/O pins	14 (of which six provide PWM output)
Analogue input pins	6
DC current per I/O pin	40 mA
DC current for 3.3V pin	50 mA
Flash memory	32 kB (ATmega328), of which 0.5 kB is used by bootloader
SRAM	2 kB (ATmega328)
EEPROM	1 kB (ATmega328)
Clock speed	16 MHz

Note: Arduino Uno board has been discontinued and its new version (Arduino Uno R2) is available now. Both boards can be used for this circuit

corresponding instructions transmitted through the RF transmitter to control the robot. The robot receives the instructions and the MCU section drives

motors corresponding to instructions received. Obstacle detectors sense any object in the way and intimate the microcontroller to stop forward motion.

Figs 3 and 4 show circuits of the wireless steering and the robot, respectively.

Wireless steering. It is built around Arduino Uno board (Board1), encoder HT12E (IC1), regulator 7805 (IC2), 433MHz RF transmitter module and a few discrete components. Pins A0, A1, A2, A3, A4 and A5 of Board1 are connected to pins ST, Z-axis, Y-axis, X-axis, GND and V_{CC} of the accelerometer module, respectively.

The microcontroller of Board1 receives data for x, y and z axes from the accelerometer. This data is continuously compared with predefined